PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions

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March 23, 2026

Abstract

This pilot study examines whether inter-individual differences moderate susceptibility to cyber-manipulation in a high-dimensional political opinion state space. Rather than estimating a no-attack counterfactual, run_5 focuses on a narrower and methodologically cleaner question: among attacked individuals, which profile differences predict larger post-minus-baseline opinion shifts after exposure to a common adversarial misinformation vector? Using ontology-constrained scenario generation, repeated opinion leaves, structured large-language-model agents, and audited realism/coherence checks, the pipeline generated 20 attacked scenarios across 4 repeated opinion leaves in Defense_and_National_Security. The attacked sample showed a mean opinion delta of 26.6 (SD = 35.0), with mean attack realism 0.70 and mean post-exposure plausibility 0.62. In the primary robust delta model, the coefficient for the pre-registered susceptibility moderator was -16.97 ($p = 0.010$; bootstrap 95% CI [-29.78, -1.03]). The attack-only design directly matches the moderation question but does not estimate absolute attack-versus-no-attack effects. Findings are therefore methodological and exploratory rather than claim-ready evidence about real-world populations.

1 Introduction

Digitally mediated political persuasion increasingly operates through selective framing, repetition, identity cues, and misinformation-rich attention environments rather than through overtly technical intrusion. False or misleading political content can spread rapidly online, distort perceived consensus, and shape trust in institutions and policy judgement (Bennett & Livingston, 2018; Lazer et al., 2018; Lewandowsky et al., 2017; Vosoughi et al., 2018). Research on fake news and online judgement further suggests that susceptibility is not uniform across individuals; it depends on reasoning style, attention, prior beliefs, and the structure of the informational environment (Kozyreva et al., 2020; Pennycook & Rand, 2019, 2021; Pennycook et al., 2021; Roozenbeek et al., 2022).

Within security studies, these same dynamics are increasingly discussed under the headings of cognitive warfare and cognitive superiority, where the strategic target is not only information accuracy but also the shaping of attention, belief revision, emotion, and collective will. NATO-affiliated concept development has framed the cognitive domain as an expanding arena of contestation beyond land, maritime, air, cyber, and space, emphasizing societal resilience and judgement under sustained informational pressure (Atkinson, 2021; Paulauskas, 2024). Ethical analysis likewise stresses that cognitive warfare concerns interference with agency, autonomy, and institutional trust, not merely messaging efficiency (Hung & Hung, 2022; Miller, 2023).

The present study uses *cognitive sovereignty* as an analytic framing for the extent to which political judgement remains self-directed under manipulative digital exposure. We treat the term as a methodological construct rather than a settled psychometric entity. The research question is: *How do inter-individual differences moderate the effectivity of digital adversarial attacks on political opinions?* In this pilot, effectivity is operationalized as the within-individual opinion delta between a baseline judgement and a post-exposure judgement after a common attack-vector family has been applied.

This framing led to a methodological redesign. Earlier pilots mixed treated and no-attack control scenarios, which answered a somewhat different question and loaded unnecessary heterogeneity into the SEM. run_5 instead adopts an attacked-only design: every simulated individual receives the same misinformation attack-vector family, and the analysis asks which profile differences predict larger post-baseline movement. This design does not estimate the absolute incremental effect of attack versus no attack, but it aligns more directly with the moderation question of interest.

A multi-agent simulation architecture is appropriate here because direct empirical datasets that simultaneously preserve baseline state, structured profile descriptors, attack-vector semantics, and issue-specific post-exposure opinion states are difficult to obtain and often misaligned with ontology-level research needs. Agent-based social simulation has long been defended as a legitimate scientific instrument when the model structure is explicit, auditable, and treated as a generative research tool rather than a substitute for direct observation (Cioffi-Revilla, 2002). The present contribution is therefore methodological: it builds an ontology-driven backend capable of generating auditable attacked scenarios, evaluating them with structured LLM agents, and estimating moderation in a form that can later be scaled and stress-tested.

2 Materials and Methods

2.1 Design Logic

run_5 used the test ontology triplet and focused on the defense-related opinion domain (Defense_and_National_Security) to reduce issue heterogeneity while keeping the state space substantively relevant to the cognitive-warfare framing. The attack ontology was restricted to a single leaf, `ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing`, so that the pilot would estimate heterogeneity in susceptibility to a common adversarial tactic rather than conflating moderator variation with attack-family variation. Scenario generation used repeated opinion leaves rather than near-unique issue sampling, and candidate profiles were oversampled before selection so that low and high values of the pre-registered susceptibility index were both represented across the attacked sample.

Hierarchical ontologies were preserved upstream, but only leaf nodes were sampled for estimation. PROFILE included mixed variable types, combining categorical attributes such as sex with continuous percentile-based traits. OPINION leaves encoded policy-specific positions, and ATTACK leaves encoded social-media persuasion tactics rather than hacking or infrastructure disruption. Baseline and post-exposure opinions were recorded on a high-resolution signed scale from -1000 to $+1000$.

2.2 Agentic Pipeline

The pipeline proceeded in sequential stages: scenario creation, baseline opinion assessment, attack generation, post-attack opinion assessment, delta construction, moderation modeling, interactive visualization, publication-asset generation, and manuscript build. Prompting was tuned to produce realistic platform-native manipulation rather than theatrical propaganda. Generated attack exposures were reviewed by a realism agent and rewritten once when realism or coherence fell below threshold. Baseline and post-exposure opinion assessments underwent a second self-supervision pass designed to penalize coarse rounding, implausible reversals, and profile-inconsistent issue responses.

2.3 Operationalization and Model Specification

The primary moderator was fixed in advance as `profile_cont_susceptibility_index`. The primary outcome for run_5 was the attacked-only opinion delta, defined as post-exposure score minus baseline score. Because all scenarios were attacked, moderation is estimated directly as variation in delta across profiles rather than as an attack-presence interaction.

The main path model was:

$$\text{baseline}_i = \alpha_b + \beta_1 S_i + \gamma' L_i + \varepsilon_{bi} \tag{1}$$

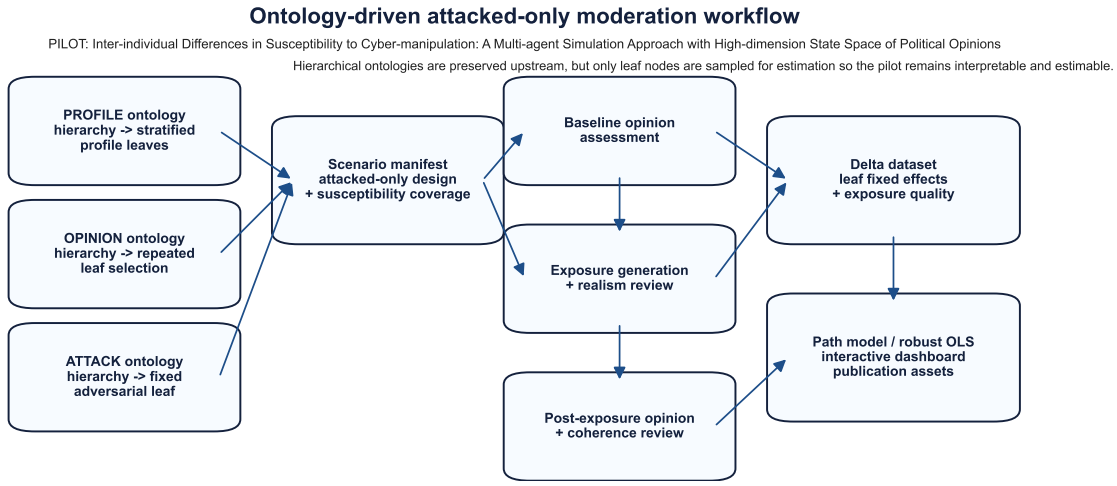
$$\Delta_i = \alpha_\Delta + \lambda_1 \text{baseline}_i + \lambda_2 \left|\text{baseline}_i\right| + \beta_2 S_i + \beta_3 Q_i + \delta' L_i + \varepsilon_{\Delta i} \tag{2}$$

where S_i denotes the standardized susceptibility moderator, Q_i denotes an exposure-quality composite derived from attack realism, coherence, and intensity, and L_i denotes opinion-leaf fixed effects. This structure treats baseline opinion as an anchor and delta as the substantive response variable. Robust OLS with HC3 standard errors and percentile bootstrap intervals were reported alongside the SEM/path model (Efron & Tibshirani, 1993; Rosseel, 2012).

3 Results

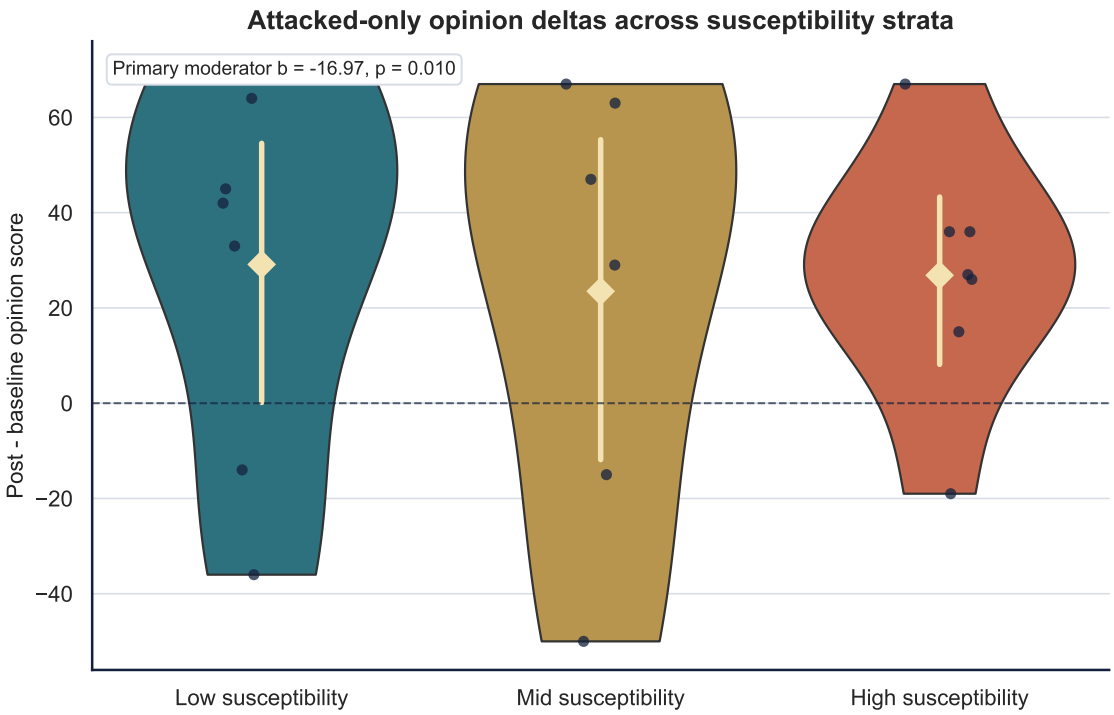
The attacked-only design produced 20 scenarios over 4 repeated opinion leaves. Mean baseline opinion was 176.2, mean post-exposure opinion was 202.8, and mean opinion delta was 26.6. Mean attack realism was 0.70, and mean post-exposure plausibility was 0.62. Figure 1 summarizes the ontology-driven workflow. Figure 2 shows attacked-only deltas stratified by susceptibility tercile. Figure 3 plots the model-implied delta trend over the primary susceptibility moderator. Figure 4 displays the annotated path model used in the main analysis.

Figure 1: Ontology-driven attacked-only study design.



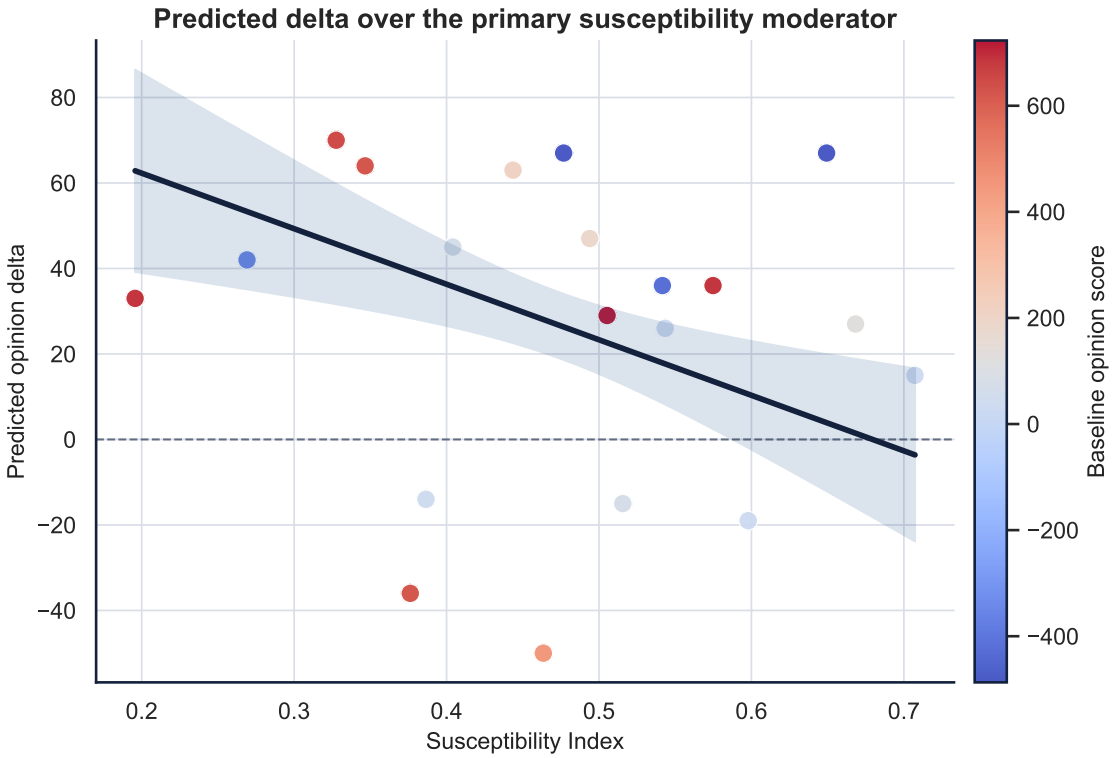
Note. The hierarchical PROFILE, ATTACK, and OPINION ontologies are preserved upstream. Only leaf nodes are sampled for estimation, and all scenarios in run_5 receive the same attack-vector family so the analysis isolates heterogeneity in post-baseline response.

Figure 2: Opinion-delta distribution across susceptibility strata.



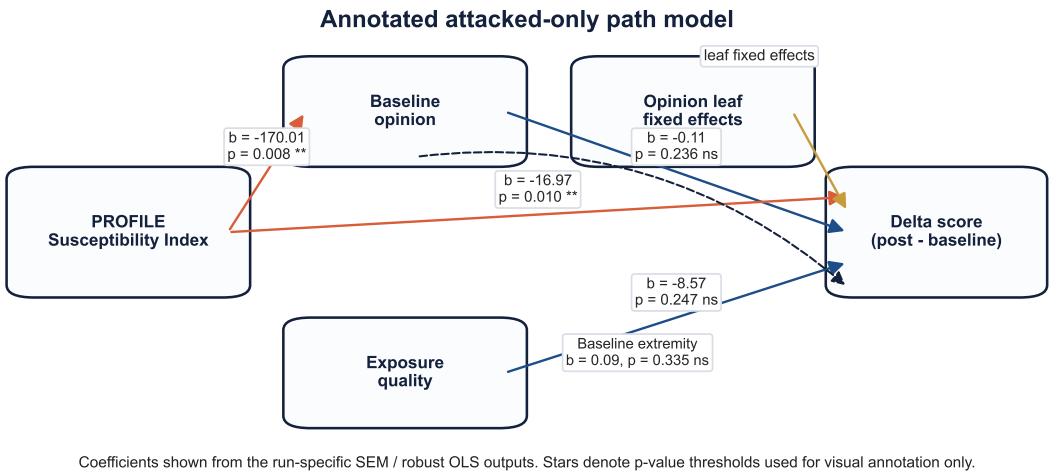
Note. Violin densities, raw scenario points, and mean markers are shown for terciles of the pre-registered susceptibility moderator. This is an attacked-only comparison, not a treatment-versus-control figure.

Figure 3: Modeled delta over the primary susceptibility moderator.



Note. The line shows model-based predicted delta with baseline anchoring, baseline extremity, exposure quality, and opinion-leaf fixed effects held at observed means. Point color encodes baseline opinion score.

Figure 4: Annotated attacked-only path model.



Note. Arrows display run-specific coefficients from the SEM / robust OLS specification. Significance stars are included only as visual aids; substantive interpretation remains exploratory because the pilot sample is small.

In the primary robust delta model, the coefficient for the pre-registered susceptibility moderator was -16.97 with p = 0.010. Because the outcome is a *signed* delta, this negative coefficient should be read as directional movement toward the lower pole of the opinion scale rather than as a simple reduction in absolute responsiveness. Baseline score remained an important anchor

Table 1: Pilot design and run configuration

Parameter	Value
Run identifier	run_5
Paper title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions
Analysis mode	treated_only
Scenarios	20
Attack ratio	1.000
Attack leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
Focus opinion domain	Defense_and_National_Security
Opinion leaves sampled	4
OpenRouter model	mistralai/mistral-small-3.2-24b-instruct
Primary moderator	profile_cont_susceptibility_index
Bootstrap samples	800

Note. Run-level configuration for the pilot simulation reported in run_5.

Table 2: Descriptive statistics by susceptibility stratum

susceptibility_group	n	baseline_mean	post_mean	delta_mean	delta_sd	exposure_quality_mean
Low susceptibility	7	335.714	364.857	29.143	39.625	0.707
Mid susceptibility	6	192.833	216.333	23.500	46.732	0.707
High susceptibility	7	2.286	29.143	26.857	25.894	0.708

Note. All scenarios in run_5 received the same attack-vector family; strata are based on terciles of the primary susceptibility moderator.

Table 3: Primary delta model with robust and bootstrapped estimates

term	estimate	std_error	p_value	conf_low_x	conf_high_x	bootstrap_mean	bootstrap_std	conf_low_y	conf_high_y	n_bootstrap_success
Intercept	-18.271	12.216	0.135	-42.213	5.672	-20.534	38.751	-55.420	-1.418	800
baseline_score	-0.113	0.096	0.236	-0.300	0.074	-0.141	0.089	-0.353	-0.032	800
baseline_abs_score	0.089	0.093	0.335	-0.092	0.271	0.117	0.095	0.019	0.309	800
primary_moderator_z	-16.968	6.549	0.010	-29.802	-4.133	-15.986	8.526	-29.775	-1.033	800
exposure_quality_z	-8.572	7.406	0.247	-23.089	5.944	-6.451	12.034	-24.974	17.129	800
opinion_leaf_fe_issue_position_taxonomy_981_defense_and_national_security_170_340_conscription_support	-9.981	14.878	0.506	-39.177	19.217	-35.731	78.330	-210.310	35.140	800
opinion_leaf_fe_issue_position_taxonomy_64_103_defense_and_national_security_0_233_defense_spending_increase_support	64.103	12.179	0.000	40.233	87.974	65.959	25.212	44.604	98.439	800
opinion_leaf_fe_issue_position_taxonomy_50_176_defense_and_national_security_47_957_strategic_autonomy_support	50.176	16.438	0.000	17.957	82.395	70.039	29.943	38.206	113.076	800

Note. The primary outcome is post-minus-baseline opinion delta. Robust standard errors are HC3. Bootstrap intervals are percentile intervals from resampled pilot fits and remain exploratory given the small sample.

(b = -0.11, p = 0.236), while baseline extremity contributed 0.09 with p = 0.335. Exposure quality contributed -8.57 with p = 0.247. The bootstrap 95% interval for the primary susceptibility coefficient was [-29.78, -1.03]. The least uncertain exploratory moderators were Resilience Index (b = 16.97, p = 0.010); Big Five Conscientiousness Competence

The SEM converged with display-capped CFI = 1.000 and RMSEA = 0.000. Because semopy can report fit indices slightly outside their theoretical bounds in very small samples, the manuscript caps those values to the interpretable $[0, 1]$ range for reporting. Even so, the attacked-only structure is more coherent than the earlier control-heavy designs. Fit should still be interpreted cautiously because the model is estimated on only 20 scenarios with repeated leaf fixed effects.

4 Discussion

run_5 is methodologically stronger than earlier pilots for three reasons. First, it now estimates the construct the paper actually cares about: which inter-individual differences are associated with larger post-baseline opinion movement after a common adversarial exposure. Second, it narrows attack-side variance by fixing the attack-vector family while preserving profile and issue variation. Third, it expresses the main result directly on the delta scale rather than embedding the question in a treatment-control interaction architecture that was not central to the stated research aim.

This redesign improves interpretability but does not erase the core limitations of the pilot. The attacked-only design means the study does not estimate the incremental effect of attack relative to a no-attack counterfactual. Instead, it estimates heterogeneity of response among attacked individuals. That is the correct target for the present pilot, but it also means the findings should not be presented as causal population effects of cyber-manipulation in the wild. In addition, the primary coefficient is estimated on a signed delta scale, so moderator direction should not be collapsed into an unsigned notion of “more” or “less” susceptibility without a companion analysis of absolute opinion movement. With only 20 scenarios, the paper should be read as workflow validation and model-design refinement rather than as substantive evidence that a specific moderator has been established.

Two additional points matter for interpretation. First, realism and coherence were explicitly audited, and the exposure-quality term was included to absorb part of the variance attributable to message quality rather than profile differences. Second, the ontology-constrained repeated-leaf design keeps the scenarios closer to interpretable policy-space comparisons than a fully unconstrained random sample would. Those design choices make the synthetic system more scientifically legible even when the sample is still small.

5 Conclusion

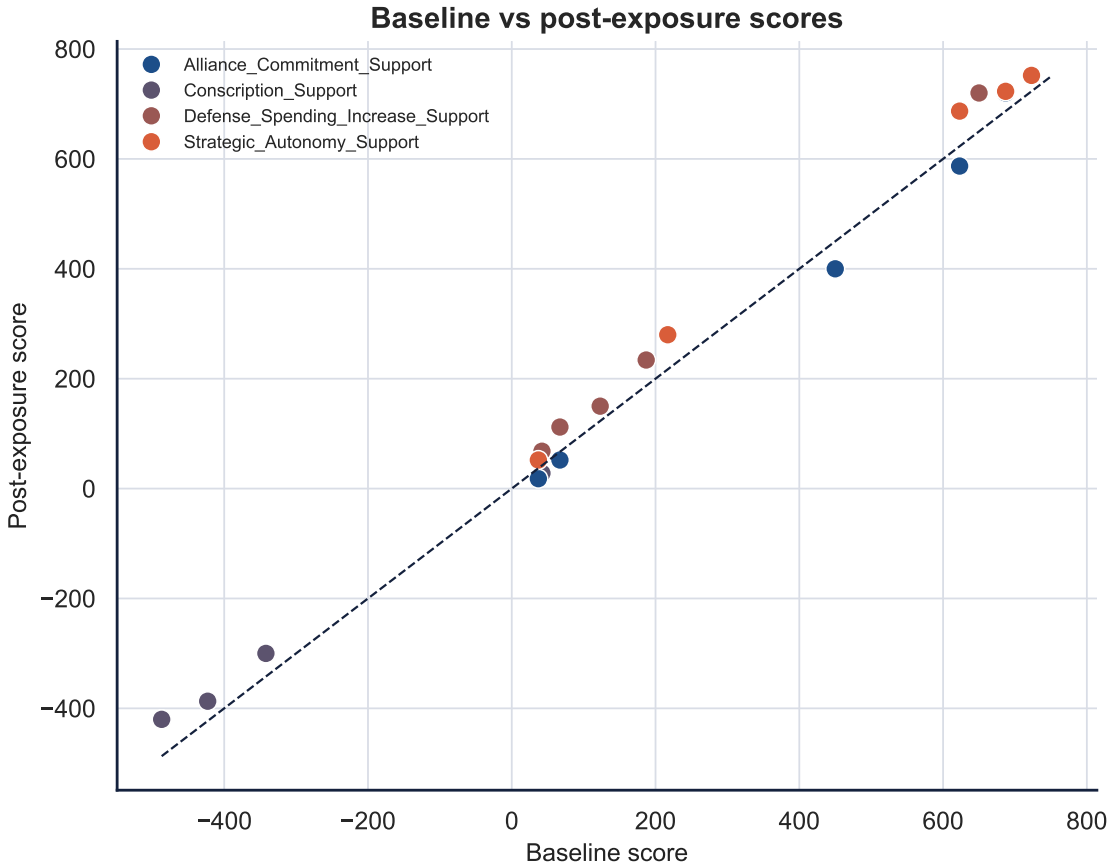
This pilot supports the feasibility of an ontology-driven multi-agent simulation framework for studying how inter-individual differences may moderate susceptibility to cyber-manipulation in political opinion spaces. In its current form, the strongest answer the paper can give is methodological: the attacked-only pipeline is operational, auditable, and substantially better aligned with the research question than the earlier treatment-control architecture. The next step is not stronger rhetoric but stronger design: larger attacked samples, multiple attack

leaves, multi-model sensitivity analysis, and eventual triangulation against empirical human data wherever ethical and legally feasible.

A Supplementary Materials

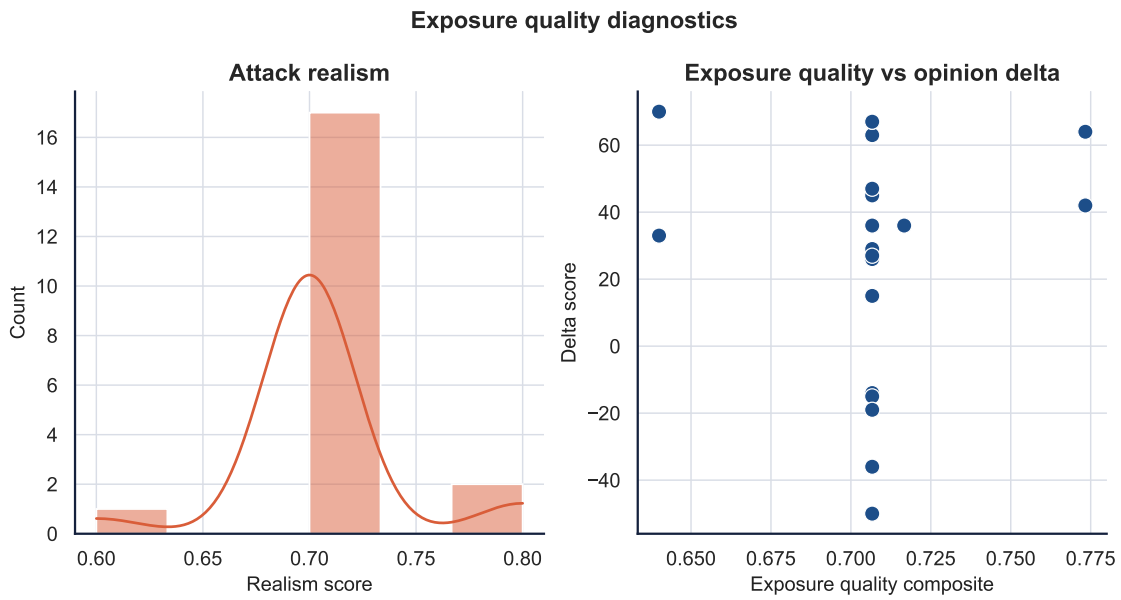
The repository includes an interactive dashboard generated from run_5. The static publication assets below are the paper-ready subset of that richer exploratory output.

Figure 5: Baseline versus post-exposure opinion scores.



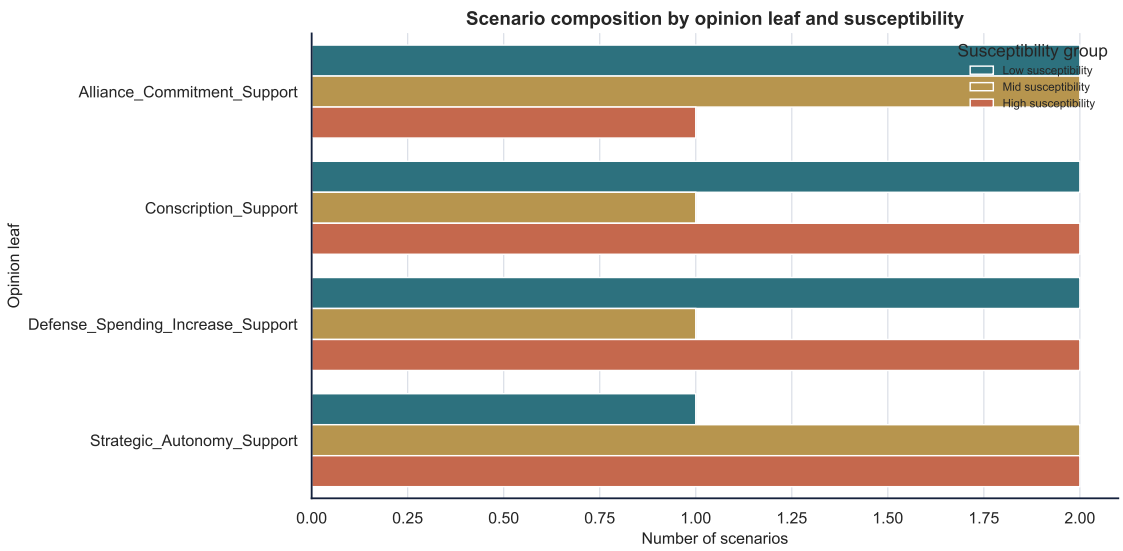
Note. The diagonal line marks no change. Points are colored by opinion leaf to show how repeated-leaf sampling constrains issue heterogeneity without collapsing the political state space to a single item.

Figure 6: Exposure-quality diagnostics.



Note. The left panel shows the realism distribution assigned by the reviewer agent. The right panel relates the exposure-quality composite to the observed opinion delta.

Figure 7: Scenario composition by opinion leaf and susceptibility.



Note. Repeated opinion leaves were combined with susceptibility-stratified profile sampling so that the pilot can estimate moderation without dispersing all observations across unique items.

Figure 8: Primary and exploratory moderator coefficients.

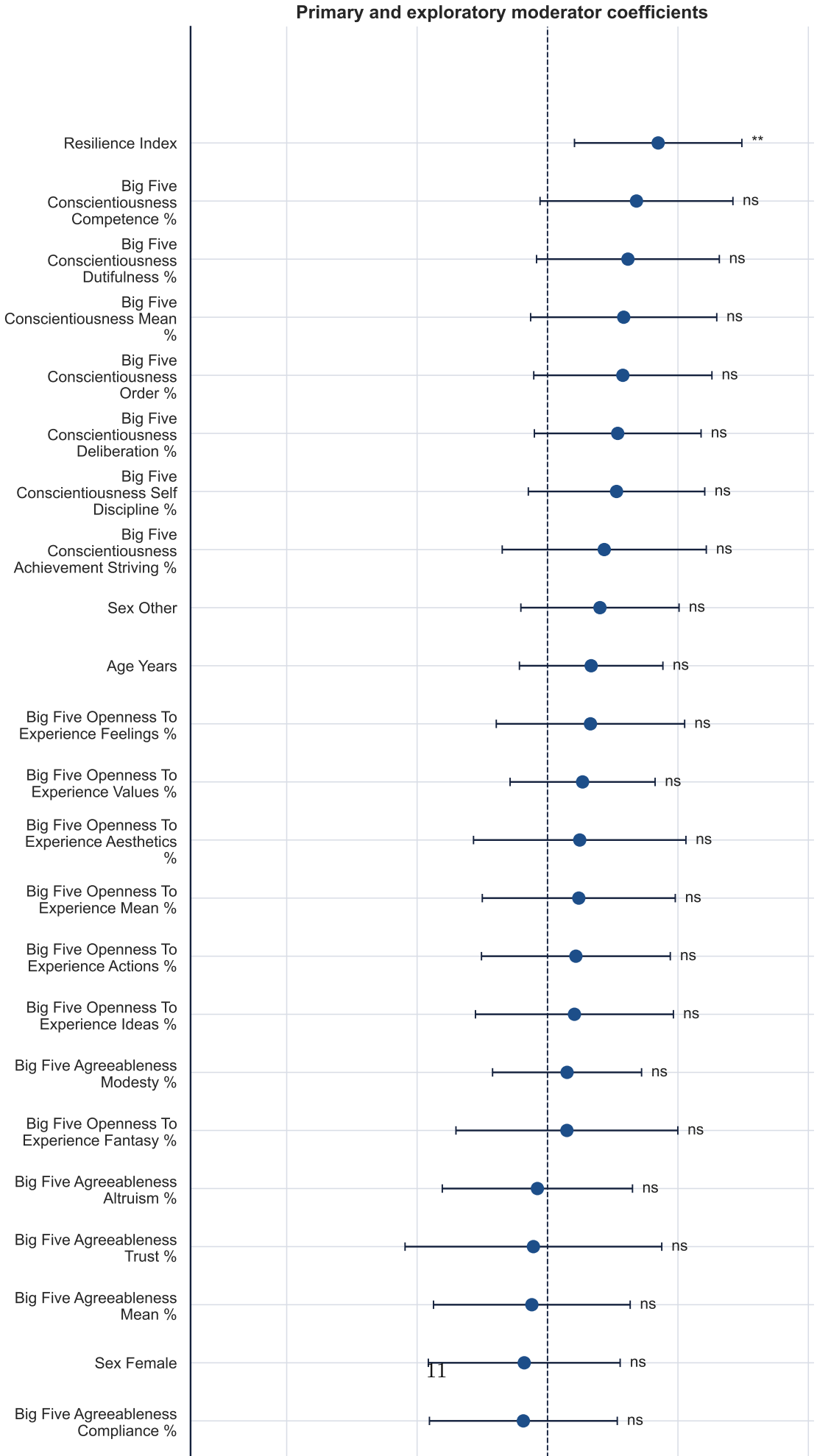


Table 4: Ontology leaves used in the pilot

ontology	leaf_label	leaf_path	n_scenarios
OPINION	Alliance_Commitment_Support	Issue_Position_Taxonomy > Defense_and_National_Security > Alliance_Commitment_Support	5
OPINION	Conscription_Support	Issue_Position_Taxonomy > Defense_and_National_Security > Conscription_Support	5
OPINION	Defense_Spending_Increase_Support	Issue_Position_Taxonomy > Defense_and_National_Security > Defense_Spending_Increase_Support	5
OPINION	Strategic_Autonomy_Support	Issue_Position_Taxonomy > Defense_and_National_Security > Strategic_Autonomy_Support	5
ATTACK	Fixed attack leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing	20

Note. Opinion leaves reflect the scenario allocation in run_5; the attack ontology was restricted to one common misinformation vector for this attacked-only pilot.

Table 5: Exploratory moderator comparison

moderator_column	moderator_label	role	effect_kind	effect_estimate	effect_std_error	effect_p_value	effect_conf_low	effect_conf_high	moderator_main_estimate	moderator_main_p_value	interaction_estimate	interaction_std_error	interaction_p_value	interaction_conf_low	interaction_conf_high
profile_cont_resilience_index	Resilience Index	exploratory	moderator_main_effect	16.968	6.549	0.010	4.133	29.802	16.968	0.010	16.968	6.549	0.010	4.133	29.802
profile_cont_big_five_conscientiousness_competence_pct	Big Five Conscientiousness Competence %	exploratory	moderator_main_effect	13.642	7.546	0.071	-1.147	28.432	13.642	0.071	13.642	7.546	0.071	-1.147	28.432
profile_cont_big_five_conscientiousness_dutifulness_pct	Big Five Conscientiousness Dutifulness %	exploratory	moderator_main_effect	12.332	7.150	0.085	-1.682	26.345	12.332	0.085	12.332	7.150	0.085	-1.682	26.345
profile_cont_big_five_conscientiousness_order_pct	Big Five Conscientiousness Order %	exploratory	moderator_main_effect	11.550	6.970	0.098	-2.112	25.211	11.550	0.098	11.550	6.970	0.098	-2.112	25.211
profile_cont_big_five_conscientiousness_deliberation_pct	Big Five Conscientiousness Deliberation %	exploratory	moderator_main_effect	10.765	6.522	0.099	-2.018	23.548	10.765	0.099	10.765	6.522	0.099	-2.018	23.548
profile_cont_big_five_conscientiousness_mean_pct	Big Five Conscientiousness Mean %	exploratory	moderator_main_effect	11.681	7.282	0.109	-2.592	25.954	11.681	0.109	11.681	7.282	0.109	-2.592	25.954
profile_cont_big_five_conscientiousness_self_discipline_pct	Big Five Conscientiousness Self Discipline %	exploratory	moderator_main_effect	10.584	6.902	0.125	-2.944	24.112	10.584	0.125	10.584	6.902	0.125	-2.944	24.112
profile_cat__profile_cat_sex_Other	Sex Other	exploratory	moderator_main_effect	8.041	6.189	0.194	-4.088	20.171	8.041	0.194	8.041	6.189	0.194	-4.088	20.171
profile_cont_big_five_neuroticism_anger_hostility_pct	Big Five Neuroticism Anger Hostility %	exploratory	moderator_main_effect	-9.336	7.729	0.227	-24.485	5.813	-9.336	0.227	-9.336	7.729	0.227	-24.485	5.813
profile_cont_age_years	Age Years	exploratory	moderator_main_effect	6.692	5.613	0.233	-4.310	17.693	6.692	0.233	6.692	5.613	0.233	-4.310	17.693
profile_cont_big_five_neuroticism_anxiety_pct	Big Five Neuroticism Anxiety %	exploratory	moderator_main_effect	-6.961	5.895	0.238	-18.516	4.594	-6.961	0.238	-6.961	5.895	0.238	-18.516	4.594
profile_cont_big_five_conscientiousness_achievement_striving_pct	Big Five Conscientiousness Achievement Striving %	exploratory	moderator_main_effect	8.701	7.984	0.276	-6.948	24.349	8.701	0.276	8.701	7.984	0.276	-6.948	24.349
profile_cont_big_five_extroversion_warmth_pct	Big Five Extraversion Warmth %	exploratory	moderator_main_effect	-15.452	14.420	0.284	-43.715	12.810	-15.452	0.284	-15.452	14.420	0.284	-43.715	12.810
profile_cont_big_five_extroversion_assertiveness_pct	Big Five Extraversion Assertiveness %	exploratory	moderator_main_effect	-13.080	12.326	0.289	-37.238	11.079	-13.080	0.289	-13.080	12.326	0.289	-37.238	11.079
profile_cont_big_five_neuroticism_impulsiveness_pct	Big Five Neuroticism Impulsiveness %	exploratory	moderator_main_effect	-7.217	7.043	0.306	-21.021	6.588	-7.217	0.306	-7.217	7.043	0.306	-21.021	6.588
profile_cont_big_five_neuroticism_mean_pct	Big Five Neuroticism Mean %	exploratory	moderator_main_effect	-7.201	7.397	0.330	-21.699	7.298	-7.201	0.330	-7.201	7.397	0.330	-21.699	7.298
profile_cont_big_five_openness_to_experience_values_pct	Big Five Openness To Experience Values %	exploratory	moderator_main_effect	5.377	5.674	0.343	-5.744	16.498	5.377	0.343	5.377	5.674	0.343	-5.744	16.498
profile_cat__profile_cat_sex_Male	Sex Male	exploratory	moderator_main_effect	-6.819	7.343	0.353	-21.211	7.574	-6.819	0.353	-6.819	7.343	0.353	-21.211	7.574
profile_cont_big_five_openness_to_experience_feelings_pct	Big Five Openness To Experience Feelings %	exploratory	moderator_main_effect	6.584	7.379	0.372	-7.860	21.029	6.584	0.372	6.584	7.379	0.372	-7.860	21.029
profile_cont_big_five_extroversion_excitement_seeking_pct	Big Five Extraversion Excitement Seeking %	exploratory	moderator_main_effect	-12.204	14.435	0.398	-40.496	16.088	-12.204	0.398	-12.204	14.435	0.398	-40.496	16.088
profile_cont_big_five_neuroticism_self_consciousness_pct	Big Five Neuroticism Self Consciousness %	exploratory	moderator_main_effect	-6.794	8.112	0.403	-22.683	9.115	-6.794	0.403	-6.794	8.112	0.403	-22.683	9.115
profile_cont_big_five_neuroticism_stress_vulnerability_pct	Big Five Neuroticism Stress Vulnerability %	exploratory	moderator_main_effect	-5.865	7.366	0.426	-20.303	8.572	-5.865	0.426	-5.865	7.366	0.426	-20.303	8.572
profile_cont_big_five_extroversion_mean_pct	Big Five Extraversion Mean %	exploratory	moderator_main_effect	-12.011	15.647	0.443	-42.679	18.657	-12.011	0.443	-12.011	15.647	0.443	-42.679	18.657
profile_cont_big_five_extroversion_positive_emotions_pct	Big Five Extraversion Positive Emotions %	exploratory	moderator_main_effect	-9.985	13.425	0.457	-36.298	16.327	-9.985	0.457	-9.985	13.425	0.457	-36.298	16.327
profile_cont_big_five_agreeableness_tender_mindedness_pct	Big Five Agreeableness Tender Mindedness %	exploratory	moderator_main_effect	-5.371	7.799	0.486	-20.480	9.738	-5.371	0.486	-5.371	7.799	0.486	-20.480	9.738
profile_cont_big_five_extroversion_activity_level_pct	Big Five Extraversion Activity Level %	exploratory	moderator_main_effect	-10.876	16.515	0.510	-43.245	21.493	-10.876	0.510	-10.876	16.515	0.510	-43.245	21.493
profile_cont_big_five_openness_to_experience_mean_pct	Big Five Openness To Experience Mean %	exploratory	moderator_main_effect	4.793	7.549	0.525	-10.003	19.590	4.793	0.525	4.793	7.549	0.525	-10.003	19.590
profile_cont_big_five_openness_to_experience_aesthetics_pct	Big Five Openness To Experience Aesthetics %	exploratory	moderator_main_effect	4.938	8.310	0.552	-11.350	21.226	4.938	0.552	4.938	8.310	0.552	-11.350	21.226
profile_cont_big_five_openness_to_experience_actions_pct	Big Five Openness To Experience Actions %	exploratory	moderator_main_effect	4.349	7.387	0.556	-10.129	18.826	4.349	0.556	4.349	7.387	0.556	-10.129	18.826
profile_cont_big_five_openness_to_experience_ideas_pct	Big Five Openness To Experience Ideas %	exploratory	moderator_main_effect	4.127	7.744	0.594	-11.051	19.306	4.127	0.594	4.127	7.744	0.594	-11.051	19.306
profile_cont_big_five_agreeableness_modesty_pct	Big Five Agreeableness Modesty %	exploratory	moderator_main_effect	2.997	5.830	0.607	-8.429	14.422	2.997	0.607	2.997	5.830	0.607	-8.429	14.422
profile_cont_big_five_agreeableness_compliance_pct	Big Five Agreeableness Compliance %	exploratory	moderator_main_effect	-3.706	7.349	0.614	-18.110	10.697	-3.706	0.614	-3.706	7.349	0.614	-18.110	10.697
profile_cat__profile_cat_sex_Female	Sex Female	exploratory	moderator_main_effect	-3.571	7.594	0.634	-18.278	11.137	-3.571	0.634	-3.571	7.594	0.634	-18.278	11.137
profile_cont_big_five_neuroticism_depression_pct	Big Five Neuroticism Depression %	exploratory	moderator_main_effect	-3.852	8.215	0.639	-19.954	12.250	-3.852	0.639	-3.852	8.215	0.639	-19.954	12.250
profile_cont_big_five_agreeableness_straightforwardness_pct	Big Five Agreeableness Straightforwardness %	exploratory	moderator_main_effect	-3.738	8.027	0.641	-19.470	11.995	-3.738	0.641	-3.738	8.027	0.641	-19.470	11.995
profile_cont_big_five_openness_to_experience_fantasy_pct	Big Five Openness To Experience Fantasy %	exploratory	moderator_main_effect	2.970	8.681	0.732	-14.043	19.984	2.970	0.732	2.970	8.681	0.732	-14.043	19.984
profile_cont_big_five_agreeableness_mean_pct	Big Five Agreeableness Mean %	exploratory	moderator_main_effect	-2.405	7.694	0.755	-17.485	12.675	-2.405	0.755	-2.405	7.694	0.755	-17.485	12.675
profile_cont_big_five_extroversion_gregariousness_pct	Big Five Extraversion Gregariousness %	exploratory	moderator_main_effect	-4.891	15.793	0.757	-35.844	26.062	-4.891	0.757	-4.891	15.793	0.757	-35.844	26.062
profile_cont_big_five_agreeableness_trust_pct	Big Five Agreeableness Trust %	exploratory	moderator_main_effect	-2.165	10.043	0.829	-21.849	17.519	-2.165	0.829	-2.165	10.043	0.829	-21.849	17.519
profile_cont_big_five_agreeableness_altruism_pct	Big Five Agreeableness Altruism %	exploratory	moderator_main_effect	-1.554	7.441	0.835	-16.138	13.030	-1.554	0.835	-1.554	7.441	0.835	-16.138	13.030
profile_cont_susceptibility_index	Susceptibility Index	primary	moderator_main_effect	-16.968	6.549	0.010	-29.802	-4.133	-16.968	0.010	-16.968	6.549	0.010	-29.802	-4.133

Note. Each moderator was estimated in a separate delta model with baseline anchoring, exposure-quality control, and opinion fixed effects included as controls.

Table 6: Assumption and peer-review risk register

section	item	status	detail
Assumption register	Pilot sample size is sufficient for stable moderation estimates.	risk	Treat estimates as exploratory and scale scenario count in later runs.
Assumption register	Attack-only design is intentional and aligned with the moderation question.	ok	Interpret results as moderation among attacked individuals, not as a no-attack counterfactual contrast.
Assumption register	Opinion measurement has enough numeric resolution.	ok	Prompt enforces high-resolution scores in [-1000,1000].
Assumption register	Attack text realism is adequate.	ok	Self-supervised realism reviewer + rewrite loop in Stage 03.
Assumption register	SEM converges with acceptable fit.	ok	Use robust OLS supplement, reduce heterogeneity, and increase scenario count before substantive interpretation.
Assumption register	LLM outputs are reproducible and auditable.	ok	Persist all prompts/outputs in provenance/raw_llm.
Peer-review critique	Synthetic LLM agents may not represent human cognition or causal response behavior.	addressed	Added transparent assumption register and explicit exploratory caveat in reports.
Peer-review critique	Adversarial content may be unrealistic or too generic.	addressed	Added realism review agent with rewrite loop and heuristic checks.
Peer-review critique	Model dependence on a single LLM could bias outputs.	addressed	Model is CLI-configurable and run metadata captures exact model for replication.
Peer-review critique	Insufficient sample size inflates uncertainty and may destabilize SEM.	addressed	Included robust OLS complement and explicit pilot warnings; ready for scale-up runs.
Peer-review critique	Attack-only designs cannot estimate the incremental effect of exposure relative to a no-attack counterfactual.	addressed	The report now states explicitly that run_5 estimates heterogeneity of post-minus-baseline response among attacked individuals only.

Note. This table summarizes the principal methodological assumptions and the explicit mitigations implemented in the pilot pipeline.

Table 7: Reproducibility manifest

Field	Value
run_id	run_5
output_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/evaluation/run
ontology_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/src/backend/o
attack_leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
openrouter_model	mistralai/mistral-small-3.2-24b-instruct
primary_moderator	profile_cont_susceptibility_index
opinion_leaf_count	38
attack_leaf_count	9

Note. Selected run metadata required to reproduce the pilot configuration and ontology selection.

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